16 y is directly proportional to $\sqrt[3]{x}$

$y = 1\frac{1}{6}$ when $x = 8$

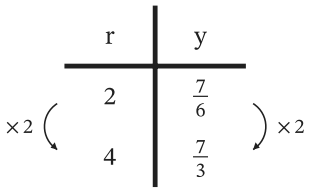
Find the value of y when $x = 64$

Let r be the cube root of x. Then y = k r.

When x = 8, r = 2: $\frac{7}{6} = 2k$ $k = \frac{7}{12}$

1 mark
Use direct proportion to the cube root

1 mark
Find the constant k



When x = 64, r = 4.

$y = \frac{7}{12} \times 4 = \frac{7}{3}$

1 mark
Correct exact value

$\frac{7}{3}$

(Total for Question 16 is 3 marks)

17 n is an integer.

Prove algebraically that the sum of $\frac{1}{2}n(n+1)$ and $\frac{1}{2}(n+1)(n+2)$ is always a square number.

(Total for Question 17 is 2 marks)

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